Advanced Macroeconomics: Computational Methods in Macroeconomics

Lecture 4

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Heterogeneous-Agent Models and Aggregate Uncertainty

- So far, we have focused on economies with agent (ex-post) heterogeneity and idiosyncratic risk
- We will now go over different algorithms to solve model with **aggregate uncertainty**
- Solving models with aggregate uncertainty is significantly more challenging
  - The extra difficulty comes from the fact that agents need to predict future prices in order to make their dynamic decisions
  - But those prices are a function of the whole distribution of agents
- As stated by KS 1998:
  - "An important component of our analysis involves dealing with the main computational difficulty of dynamic heterogeneous-agent models: in order to predict prices, consumers need to keep track of the evolution of the (wealth) distribution"
- The issue is that the distribution is an infinite-dimensional object (there is a continuum of agents)
- We need to make some assumptions (or impose some restrictions) to solve these problems

Krusell-Smith, 1997 and 1998

Krusell-Smith 1998

- To illustrate things, let's consider the toy model in Krusell-Smith 1998
- Consumers are heterogeneous in 2 dimensions: employment status $\epsilon = \{0, 1\}$ and assets
- There is an aggregate shock $z = \{z_b, z_g\}$ that affects the productivity of the economy (bad; good)
- The idiosyncratic ϵ and the aggregate shock z are correlated: more people are unemployed in bad times
- The aggregate production function is:

$$Y_t = F(z_t, K_t, L_t) = z_t K^{\alpha} L^{(1-\alpha)}$$

- The resource constraint of the economy is $C_t + K_{t+1} - (1-\delta)K_t = Y_t$

Krusell-Smith 1998

- Capital serves 2 purposes: production and store of value
- **Markets are incomplete**: households can only save in k
- We can write households' recursive problem as follows:

$$V(k, \epsilon, z, \Omega) = \max_{c,k'} u(c) + \beta \mathbb{E}\left[V\left(k', \epsilon', z', \Omega'\right)\right]$$

s.t.

$$c + k' = r(\Omega, z)k + w(\Omega, z)\epsilon + (1-\delta)k$$

$$k' \geq 0$$

$$\Omega' = H\left(\Omega, z, z'\right) \text{ [conjectured law of motion]}$$

- where $r(.)$ and $w(.)$ denotes the capital rental rate and the real wage, respectively. Ω is the households' distribution (across their wealth k and employment status ϵ)
- We cannot solve this problem as it is, since the Ω is an infinite-dimensional state

Krusell-Smith 1998

- Households care about the current and future distribution Ω because that distribution affects prices, r and ω.
- From the firms' FOCs: $r(\Omega, z) = F_K(z, K, L)$ and $w(\Omega, z) = F_L(z, K, L)$
- From the market clearing conditions:

$$L = \mathbb{E}\epsilon \quad \text{(there is no disutility for labor)}$$

$$K' = \int k'(k, \epsilon, z, \Omega)\Omega(k, \epsilon)\, \mathrm{d}k\, \mathrm{d}\epsilon$$

- In this simple problem, if we know $\{K, K'\}$ then we can pin-down all current and next-period prices $\rightarrow$ we can solve the households' problem
  - $r(.) = F_K(z, K, L) = \alpha K^{\alpha-1} L^{1-\alpha} = \alpha K^{\alpha-1}$
  - $w(.) = F_L(z, K, L) = (1-\alpha) K^{\alpha} L^{-\alpha} = (1-\alpha) K^{\alpha}$
- The question is, how is K' determined? Does K' depend on higher-order moments of Ω?

Krusell-Smith 1998

- **Key step**: Assume that agents are boundedly rational in their perceptions of how Ω evolves over time.
  - The goal is to have a conjectured LOM $\hat{H}$ that is just "good enough" to track the true LOM for Ω
- For example, agents only keep track of the first I moments, $\boldsymbol{m} \equiv \{m_1, m_2, ...m_I\}$ of Ω and use those moments to conjecture the next-period Ω'
  - Those I moments will be part of our (simplified) state space $\rightarrow$ we don't need to have Ω
- We thus need to guess a parameterized functional form for $\hat{H}$ and its parameters

Krusell-Smith 1998

- Using our $\hat{H}$, we can solve the households' problem:

$$V\left(k,\epsilon,z,\boldsymbol{m}\right)=\max_{c,k'}\ u(c)+\beta\mathbb{E}\left[V\left(k',\epsilon',z',\boldsymbol{m}'\right)\right]$$

s.t.

$$c+k'=r(\boldsymbol{m},z)k+w\left(\boldsymbol{m},z\right)\epsilon+(1-\delta)k$$

$$k'\geq 0$$

$$\boldsymbol{m}'=\hat{H}\left(\boldsymbol{m},z,z'\right)$$

- Finding the right set of moments $\boldsymbol{m}$ and a good $\hat{H}$ is and art...

Krusell-Smith 1998 - Near Aggregation Result

- In KS 1998, their main finding is that keeping track only of K is enough
- They set $\boldsymbol{m} = K$ and guess the following law of motions:

$$\log K' = a_0 + a_1 \log K \text{ if } z = z_g$$

$$\log K' = b_0 + b_1 \log K \text{ if } z = z_b$$

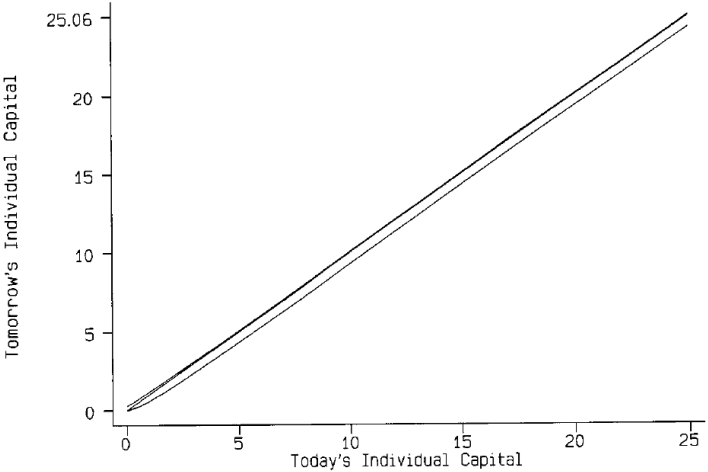
- They found that this simple approximation works extremely well $\rightarrow$ the R^2 of the previous regressions is 0.999998
- This result is usually refer to as **"near aggregation"** $\rightarrow$ we only care about the aggregate stock of capital in order to predict K'. Other moments of the distribution (volatility, kurtosis,...) do not improve the fit

Krusell-Smith 1998 - Near Aggregation Result

- **Why is only the mean important?**
- Suppose that all agents have the same marginal propensity to save
  - In this case, an additional unit of k to a low wealth household, would lead to the same increase in k' compared to a case in which I give an additional unit of k to the wealthiest household
  - But then, we don't care at all about the distribtuion $\rightarrow$ changes in that distribution (for instance, a larger wealth inequality) would not have any effects on K'.
  - All that we care about is the aggregate K

Krusell-Smith 1998 - Near Aggregation Result

- It turns out that, in the simple KS 1998 model, the policy functions of the agents are "almost" linear in their own wealth k



Krusell-Smith 1998 - Near Aggregation Result

- **Why are the marginal savings propensities almost independent of households' wealth, k?**
- This is a result that goes back to Lucas (1987)
- The utility costs from accepting fluctuations in consumption are very small, even if those fluctuations are large...
- Even though markets are incomplete and households cannot perfectly insurance against those risks, having at least 1 asset provides agents with very good insurance in utility terms.

Krusell-Smith 1998 - Full Algorithm

- Assume that agents only keep track of a finite set of moments, $\boldsymbol{m}$
- Guess a conjectured law of motion $\hat{H}(\boldsymbol{m})$. In KS 1998, we have

$$\log K' = \begin{cases} a_0 + a_1 \log K & \text{if } z = z_g \\ b_0 + b_1 \log K & \text{if} z = z_b \end{cases}$$

- Solve the households' problem using the same techniques that we have studied before

Krusell-Smith 1998 - Full Algorithm

- Using the policy functions, simulate the economy
  - KS 1998 use stochastic simulations $\rightarrow$ they generate an economy with 5000 agents, which is time consuming and may induce inaccuracies
  - Instead, you should use non-stochastic simulations, Young 2010
- Using the simulated data, run the following regressions:

$$\log K_{t+1} = \hat{a}_0 + \hat{a}_1 \log K_t + e_{t+1} \text{ if } z = z_g$$

$$\log K_{t+1} = \hat{b}_0 + \hat{b}_1 \log K_t + \nu_{t+1} \text{ if } z = z_b$$

- Use these estimates, to update the conjectured LOM
- Continue until $|\hat{a}_i - a_i| < \epsilon$ and $|\hat{b}_i - b_i| < \epsilon$
- Upon convergence, we need to check if our $\hat{H}(\boldsymbol{m})$ does a good job in tracking the observed $\{K'\}$
  - For instance, we can check whether the R^2 of our regressions is close to 1

Krusell-Smith 1997 - Model with Portfolio Choice

- Now, let's consider an extension with 2 assets: aggregate capital k and a riskless bond b
- The rest of the setup is the same to the one above
- Let's denote a household's wealth as ω

$$\omega = \left[r(\Omega, z) + (1-\delta)\right] k + b + w\left(\Omega, z\right)$$

- The households' recursive problem can be written as:

$$V\left(\omega, \epsilon, z, \Omega\right) = \max_{c, k', b'} \; u(c) + \beta \mathbb{E}\left[V\left(\omega', \epsilon', z', \Omega'\right)\right]$$

s.t.

$$c + k' + q(\Omega, z) b' = \omega$$
$$\omega' = \left[r(\Omega', z') + (1-\delta)\right] k' + b' + w\left(\Omega', z'\right)$$
$$\{k', b'\} \geq \{0, \underline{b}\}$$
$$\Omega' = H\left(\Omega, z, z'\right) \quad \text{[conjectured law of motion]}$$

Krusell-Smith **1997** - Model with Portfolio Choice

- There are now 3 market clearing conditions:

$$
\begin{aligned}
L =& \mathbb{E}\epsilon \quad \text{(there is no disutility for labor)} \\
K' =& \int k'(\omega, \epsilon, z, \Omega)\Omega(\omega, \epsilon)\, \mathrm{d}\omega\, \mathrm{d}\epsilon \\
0 =& \int b'(\omega, \epsilon, z, \Omega)\Omega(\omega, \epsilon)\, \mathrm{d}\omega\, \mathrm{d}\epsilon
\end{aligned}
$$

- Notice that knowing K is enough to pin down $r(.)$ and $w(.)$ (just as in our previous model), but it is still not enough to determine the price for the riskless bond, $q(.)$

Krusell-Smith 1997 - Model with Portfolio Choice

- Based on KS 1998, we can summarize the distribution with $\boldsymbol{m}$ moments and guess a parameterized law of motion $\hat{H}$.
- KS 1997 first considered the following approximation:

$$
\begin{aligned}
V(\omega, \epsilon, z, \boldsymbol{m}) = \max_{c,k',b'} \; & u(c) + \beta \mathbb{E}\left[V\left(\omega', \epsilon', z', \boldsymbol{m}\right)\right] \\
& c + k' + q(\boldsymbol{m}, z) b' = \omega \\
& \omega' = \left[r(\boldsymbol{m}', z') + (1-\delta)\right] k' + b' + w\left(\boldsymbol{m}', z'\right) \\
& \{k', b'\} \geq \{0, \underline{b}\} \\
& \boldsymbol{m}' = \hat{H}\left(\boldsymbol{m}, z, z'\right) \text{ [conjectured law of motion]}
\end{aligned}
$$

where $\boldsymbol{m} = K$ and

$$
\log K' = \begin{cases} a_0 + a_1 \log K & \text{if } z = z_g \\ b_0 + b_1 \log K & \text{if} z = z_b \end{cases}
$$

$$
q = \begin{cases} c_0 + c_1 \log K & \text{if } z = z_g \\ d_0 + d_1 \log K & \text{if} z = z_b \end{cases}
$$

Krusell-Smith **1997** - Model with Portfolio Choice

- Notice that they used the same LOM for aggregate capital (as in the previous example) and they guessed bond prices as a function of the aggregate stock of capital
  - Again, we need the guess for $q(.)$ because knowing K is not enough to pin down that price
- **Key result:** The previous algorithm does not work!
  - By having K as state, we can compute $r(.)$ directly and use that $r(.)$ to solve the households' problem $\rightarrow$ capital market clears in every period (by construction)
  - However, the market for bonds will only clear if the guessed $\{c_0, c_1, d_0, d_1\}$ coincide with the $\{\hat{c}_0, \hat{c}_1, \hat{d}_0, \hat{d}_1\}$ that we obtained by simulating the model and running OLS.
    - ★ This will happens only after our algorithm has converged...
  - Moreover, even if $\{c_0, c_1, d_0, d_1\}$ coincide with $\{\hat{c}_0, \hat{c}_1, \hat{d}_0, \hat{d}_1\}$, this only implies that **on average** the market for bonds clears $\rightarrow$ it does not imply that it clears in every single period of the simulation
    - ★ This is an issue, specially if aggregate B' follows a very persistent process $\rightarrow$ this lead situations in which the market clearing condition does not hold for many periods

Krusell-Smith **1997** - Model with Portfolio Choice

- Notice that adding another moment as part of $\boldsymbol{m}$ does not work either $\rightarrow$ there is just no closed form solution for the riskless bond price
- To guarantee market clearing in bonds, we use the following procedure $\rightarrow$ introduce a **"co-state"**
- We let $\boldsymbol{m} = \{K, q\}$
  - Obviously q is not a state per se, since it is an endogenous variable whose value we do not know at the beginning of each period.
  - We will treat q as if it were a variable known by households $\rightarrow$ together with a LOM for q', we can then solve for the households' problem:
- Note: KS 1997 do something slightly different in which they include q as a co-state only in the simulation stage. Let me first show you how I like to solve this problems...

Krusell-Smith 1997 - Model with Portfolio Choice

- **My preferred specification is this one:**

$$
\begin{aligned}
V(\omega, \epsilon, z, \boldsymbol{m}) = \max_{c,k',b'} \; & u(c) + \beta \mathbb{E}\left[V\left(\omega', \epsilon', z', \boldsymbol{m}\right)\right] \\
& c + k' + q(\boldsymbol{m}, z) b' = \omega \\
& \omega' = \left[r(\boldsymbol{m}', z') + (1-\delta)\right] k' + b' + w\left(\boldsymbol{m}', z'\right) \\
& \{k', b'\} \geq \{0, \underline{b}\} \\
& \boldsymbol{m}' = \hat{H}\left(\boldsymbol{m}, z, z'\right) \text{ [conjectured law of motion]}
\end{aligned}
$$

where the conjectured LOMs can be given by:

$$
\log K' = \begin{cases} a_0 + a_1 \log K & \text{if } z = z_g \\ b_0 + b_1 \log K & \text{if} z = z_b \end{cases}
$$

$$
q' = \begin{cases} c_0 + c_1 q + c_2 K' & \text{if } z' = z_g \\ d_0 + d_1 q + d_2 K' & \text{if} z' = z_b \end{cases}
$$

Krusell-Smith 1997 - Model with Portfolio Choice

- Since q is now a co-state, in the simulation stage we can solve in each period for the value of q such that the bond market clears
- This is quite easy to do... we just need to interpolate the $b'(\omega, \epsilon, z, K, q)$ policy, aggregate it using the distribution at time t, Ω_t, and solve for q such that $B' = 0$.

Krusell-Smith **1997** - Model with Portfolio Choice

- **KS 1997 do something slightly different in which they include q as a co-state only in the simulation stage**
- First, they solve this (idem as before)

$$
\begin{aligned}
V(\omega, \epsilon, z, K) = \max_{c,k',b'} \; & u(c) + \beta \mathbb{E}\left[V\left(\omega', \epsilon', z', K\right)\right] \\
& c + k' + q(K,z)b' = \omega \\
& \omega' = \left[r(K',z') + (1-\delta)\right]k' + b' + w\left(K',z'\right) \\
& \{k', b'\} \geq \{0, \underline{b}\} \\
& K' = \hat{H}\left(K', z, z'\right) \;\; \text{[conjectured law of motion]}
\end{aligned}
$$

where the conjectured LOMs are:

$$
\log K' = \begin{cases} a_0 + a_1 \log K & \text{if } z = z_g \\ b_0 + b_1 \log K & \text{if} z = z_b \end{cases}
$$

$$
q = \begin{cases} c_0 + c_1 \log K & \text{if } z = z_g \\ d_0 + d_1 \log K & \text{if} z = z_b \end{cases}
$$

Krusell-Smith 1997 - Model with Portfolio Choice

- Taking $V(.)$ as given, they simulate the model as follows:
- In each period t, they solve again for policies by solving this (simpler) problem:

$$\tilde{V}(\omega, \epsilon, z, K, q) = \max_{c,k',b'} u(c) + \beta \mathbb{E}\left[V\left(\omega', \epsilon', z', K\right)\right]$$
$$c + k' + qb' = \omega$$
$$\omega' = \left[r(K', z') + (1-\delta)\right] k' + b' + w\left(K', z'\right)$$
$$\{k', b'\} \geq \{0, \underline{b}\}$$
$$K' = \hat{H}\left(K', z, z'\right) \text{ [conjectured law of motion]}$$

- They solve the previous problem on a grid for $\{q_1, ...q_N\} \rightarrow$ they find the q such that the bond market clears in period t
- They use the simulated $\{q\}$ series (in which the bond market is now in equilibrium) to update their coefficients $\{c_0, c_1, d_0, d_1\}$

Krusell-Smith 1997 - Model with Portfolio Choice

- **Some comments:**
  - The advantage of the KS 1997 approach is that the RHS value function does not directly depend on q; hence, we have 1 fewer state variable when computing expectations
  - The disadvantage is that we need to solve for new $\{b'(..., q), k'(..., q)\}$ policies in each step of the simulation stage, which is computationally costly!

- In my preferred algorithm, we directly include q as a co-state in our Bellman equation
  - Solving the model demands more time $\rightarrow$ expectations are computed also over the q dimension
  - The simulations are MUCH faster $\rightarrow$ we just need to interpolate our policy $b'(..., q)$ to find q such that the bond market clears; we don't need to solve for $b'(..., q)$ again
  - Overall, I find this method to be faster

Boppart, Krusell, Mitman 2018

(Journal of Economic Dynamics and Control)

Boppart, Krusell, Mitman

- So far, we have viewed our problem in a **recursive** fashion
  - That is, we have written value and policy functions as $X_t = X(\boldsymbol{s}, \boldsymbol{S})$, where $\boldsymbol{s}$ is a vector of idiosyncratic states and $\boldsymbol{S}$ is a vector of aggregate states, which includes an infinite-dimensional variable, the distribution Ω
- Alternatively, we can represent the same problem and its solution on the **time or sequence space**
  - In each period, agents observe a new realization of a (vector) of random shocks, ϵ_t.
  - Their policies are written as a function of the full history of shocks. That is $X_t = Y(\epsilon_t, \epsilon_{t-1}, \epsilon_{t-2}, ...)$
- At the end of the day, notice that both $X(\boldsymbol{s}, \boldsymbol{S})$ and $Y(\epsilon_t, \epsilon_{t-1}, \epsilon_{t-2}, ...)$ are infinite-dimensional objects
  - We thus need to make some assumptions or impose some restrictions to solve our problem

Boppart, Krusell, Mitman

- In KS, we view the agents' problem in a **recursive** way and tackle the dimensionality issue by summarizing the Ω distribution with a finite set of moments $\boldsymbol{m}$ and by conjecturing a LOM for $\boldsymbol{m}'$
- Instead, in BKP, we focus on the **sequence space** (at least for the aggregates)
  - In order to break the dimensionality issue associated with the sequence representation of the problem, they make **one key assumption**: the response of any variable x to a one-time unexpected (MIT) shock can be well approximated by a linear system
  - Linearity is only imposed for aggregates $\rightarrow$ method preserves all the non-linearities at the agent level with respect to idiosyncratic shocks

Boppart, Krusell, Mitman

- How this linearity assumption works? Why does it "break" the dimensionality issue?
- Suppose that we have a model with heterogeneous agents, no aggregate uncertainty and we are able to:
  - Solve for the stationary recursive equilibrium
  - Compute the deviation from this steady state, given an unexpected (MIT) shock
- For example, suppose a (deterministic) TFP shock that follows an AR(1) process:

$$\log z_t = \rho \log z_{t-1} + \epsilon_t$$

  - The shock sequence is $\{\tilde{\epsilon}\} = \{1, 0, 0, ...0\}$ (T periods)
    (it is important for what follows that we normalize the size of the shock to 1)
  - We solve for the (deterministic) transition path (just as we have already seen before in class)
  - The transitional dynamics will give us paths for any variable of interest: aggregate savings, aggregate consumption, capital, etc...
  - Let $\{\tilde{x}\} = \{\tilde{x_0}, \tilde{x_1}, ...\tilde{x}_T\}$ denote the path for the aggregate variable x

Boppart, Krusell, Mitman

- If the response of x is linear, then the effects of shocks of different sizes and in different points in time are linearly scalable and additive
  - Put it differently, with only one (finite) shock sequence $\{\tilde{\epsilon}\} = \{1, 0, 0, ...0\}$ and based on the computed $\{\tilde{x}\} = \{\tilde{x}_0, \tilde{x}_1, ...\tilde{x}_T\}$ path, we have all the information that we need
- To see this, consider a full sequence of random shocks, $\{\epsilon\} = \{\epsilon_0, \epsilon_1, ...\epsilon_T\}$
  (for now, assume same length T as the MIT shock sequence)
  - If linearity is a "good" approximation, we can compute the evolution of x_t to these sequence of shocks with the information that we already have from our MIT shock
- At time T, for instance:

$$x_T \approx \underbrace{\tilde{x}_0 \overbrace{\epsilon_T}^{\text{current shock}}}_{\text{Current response}} + \underbrace{\tilde{x}_1 \overbrace{\epsilon_{T-1}}^{\text{previous shock}}}_{\text{1-period ahead response}} + \underbrace{\tilde{x}_2 \overbrace{\epsilon_{T-2}}^{\text{2-periods ago shock}}}_{\text{2-period ahead reponse}} + ...$$

If our impulse response has T periods:

$$x_T \approx \sum_{k=0}^{T} \tilde{x}_k \times \epsilon_{T-k}$$

Boppart, Krusell, Mitman

- Notice that we can consider a sequence of random shocks of any length
- Suppose the length of the MIT shock if T; but we want to simulate data for $N > T$ periods
- Then, at time $t > 0$, our variable of interest can be approximated as:

$$x_t \approx \sum_{k=0}^{\min\{t,T\}} \tilde{x_k} \times \epsilon_{t-k}$$

  - For any t, we simply use the latest T sequence of $\{z\}$ shocks; we discard all the other $N - T$ shocks that are farther behind in time
  - This "works" because our MIT path is large enough such that $\tilde{x_k}$ goes back to the steady state after T periods (hence for any $k > T$, $\tilde{x_k} = 0$)
  - Notice that we are normalizing variables so that they are equal to 0 in the SS

Boppart, Krusell, Mitman

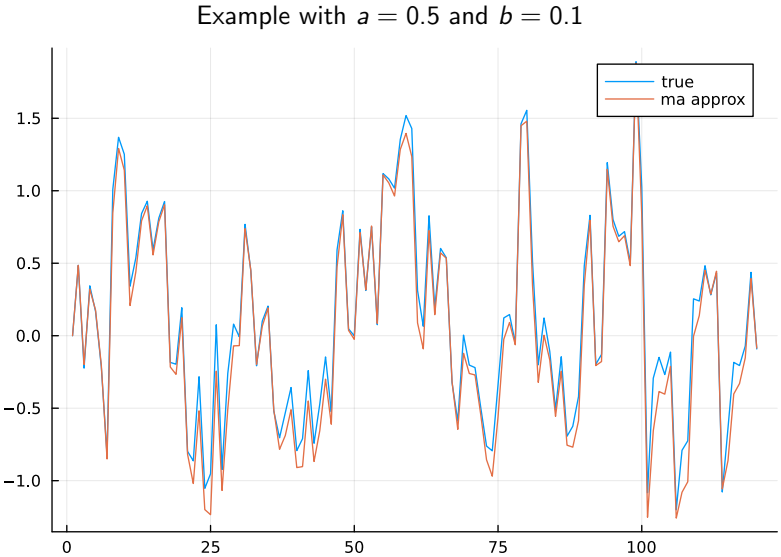
- Let's go to Julia to see how this works in practice
- Suppose that the "true" (unobserved) process for our variable of interest x is given by:

$$x_{t+1} = \alpha x_t + \beta x_t^2 + \epsilon_{t+1}$$

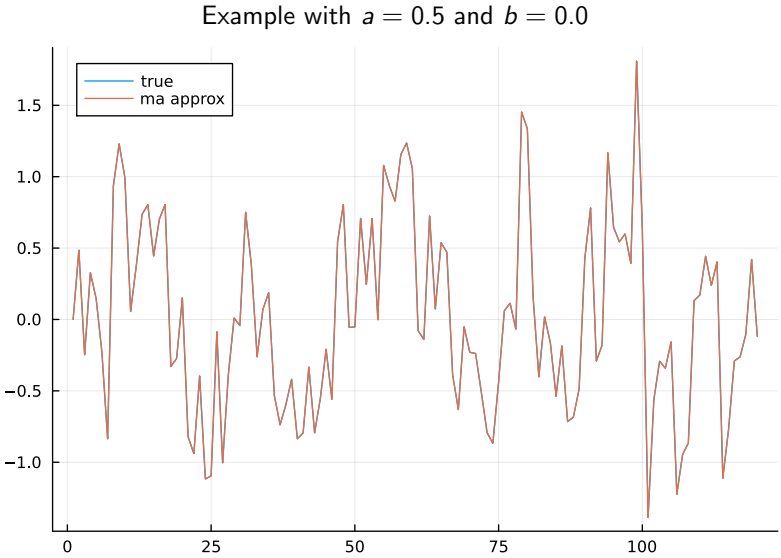
- While we do not observe that process, let's assume that we are able to compute a deterministic path to a 1-time unexpected MIT shock
  - That is, we are able to compute the evolution of x_t for the path of shocks $\{\tilde{\epsilon}\} = \{1, 0, 0, ...0\}$. We call this path $\{\tilde{x}_t\}$
- Given any sequence of shocks, $\{\epsilon\} = \{\epsilon_0, \epsilon_1, ...\epsilon_N\}$, we approximate the process as:

$$x_t \approx \sum_{k=0}^{\min\{t,T\}} \tilde{x}_k \times \epsilon_{t-k}$$

Boppart, Krusell, Mitman



Boppart, Krusell, Mitman



Boppart, Krusell, Mitman

- We can generalize our previous reasoning to account for multiple shocks
- For each shock, we need to compute a dynamic (deterministic) transition path to an MIT shock
- Notice that the size of the problem increases linearly (not exponentially)
- For instance, if we have 2 shocks $\{\epsilon, \eta\}$, then:

$$x_t \approx \sum_{k=0}^{\min\{t,N\}} \tilde{x_k}^{(\epsilon)} \times \epsilon_{t-k} + \sum_{k=0}^{\min\{t,N\}} \tilde{x_k}^{(\eta)} \times \eta_{t-k}$$

- where $\tilde{x_k}^{(\epsilon)}$ denotes the path given the ϵ shock; and $\tilde{x_k}^{(\eta)}$ denotes the path given the η shock

Boppart, Krusell, Mitman - **Full Algorithm**

- **There is really nothing new!**
- We just need to solve for the stationary recursive equilibrium of the model
- Compute transitional dynamics to an unexpected MIT shock (or shocks; depending on how many shocks we want our model to have!)
  - For this we use the backward+forward algorithm that we studied in Lecture 2
  - We guess a sequence of prices (or aggregates) and solve the agents' problem by backward induction (from $t = T$ to $t = 0$)
  - Once we have the policies for each t, we simulate forward and see if our guesses are consistent with market clearing
- Simulate a large sequence of shock (or shocks), to compute $\{x_t\}$, for every variable of interest
- We can then use the simulated $\{x_t\}$ to compute some unconditional moments or other statistics of interest

Boppart, Krusell, Mitman - Some Comments

- An important disadvantage of this method is that it does not offer a metric of fit
  - In KS, for instance, we guess a LOM $\hat{H}$, then we simulate the model and test how good that LOM is (by looking at the R^2)
  - In BKM, we cannot perform such test (at least not directly...)
- **But how do we know if linearity is a reasonable assumption?**
  - BKM provide some ideas...

1. We can consider MIT shocks of different sizes; for instance $\hat{\epsilon} = \{1, 0, ...\}$; $\hat{\epsilon} = \{2, 0, ...\}$; $\hat{\epsilon} = \{3, 0, ...\}$... We then compute impulse responses for each of those shocks and analyze if the evolution of x_t (and any other variable of interest) looks similar across them (upon re-scaling)
2. We can also compute the impulse response to an unexpected ϵ and η shock that occur at the same time. The path for x_t should be similar to the sum of the paths obtained as a response to an ϵ and to and η shock

Auclert, Bardoczy, Rognlie, Straub

The Sequence-Space Jacobian Method (Econometrica, 2021)

Introduction to SSJ

- The SSJ method provides an algorithm to compute the transitional dynamics to an unexpected MIT shock
- Using the insight from BKP, this is all we need to solve a model with aggregate uncertainty
- But wait... We already studied how to solve for transitional dynamics (Lecture 2) Why do we need the SSJ? **It's very fast!**

Introduction to SSJ

Reminder: Steps to solve for the transitional dynamics to an MIT shock at $t = 0$:

1. Solve for the steady state (before the MIT shock happens)
2. Assume that the economy is back at the initial steady after T periods
3. Guess the whole path for prices. In KS, for instance, we need to guess $\boldsymbol{r}^{guess} \equiv \left\{ r_t^{guess} \right\}_{t=1}^{T-1}$
4. Do a **backward** iteration and solve for policies in each period, $\left\{ k_{t+1}(.) \right\}_{t=1}^{T-1}$. To start the recursion, we use the fact that $V_T(.) = V_{ss}(.)$
5. Use those policies to update the distribution and **iterate forward**. For this, we use the fact that $\Omega_0(.) = \Omega_{ss}(.)$
6. Check consistency of the guess and update the guess accordingly
   - In the **KS model**, use policies and distribution to compute the path for $\left\{ K_t \right\}_{t=1}^{T-1}$ and see if the implied $\boldsymbol{r} \equiv \left\{ r_t \right\}_{t=1}^{T-1}$ (based on the firm's FOC) equals the guess
   - In the **income fluctuation problem**, we also need to check if the asset market clears but we don't have a closed-form solution for r

- **Here is the issue... How do we update our guess r^{guess}?**

Introduction to SSJ

- In the **KS model** we can set, for each $t \in [1, T-1]$, the following updating rule:

$$r_t^{update} = (1-\alpha) r_t^{guess} + \alpha r_t,$$

  where r_t is the outcome of our dynamic path and $\alpha \in (0,1)$
  - In practice, we may need to set α very close to 1... so that the updating is very small.

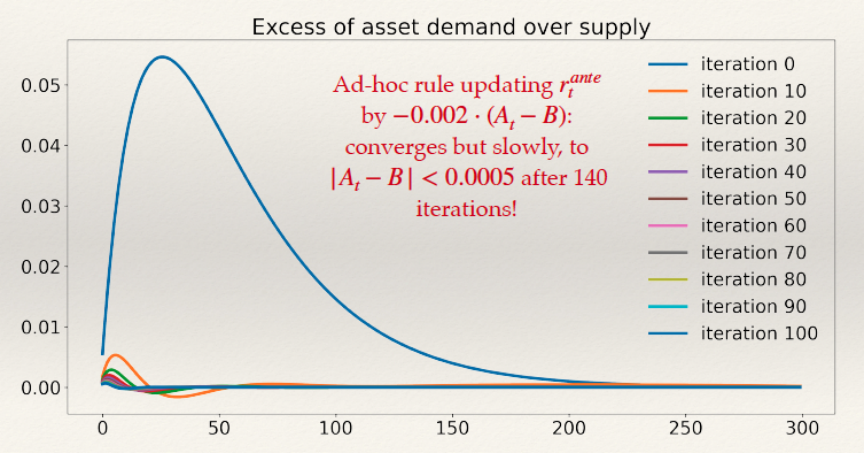
- Updating the guess is particularly an issue in the **income fluctuation problem**
  - We do not have a closed-form solution for r_t
  - To update the guess, we need to check whether markets clear and increase/decrease r_t based on whether there is excess demand/supply...
  - Check if $\mathcal{A}_{t+1}(r_t^{guess}) = \int a_{t+1}(a,y)\,\Omega_t(a,y)\,da\,dy = \bar{B}$ (with $\bar{B} = 0$ if zero net supply)
  - The updating rule for r_t can be as follows:

$$r_t^{update} = -\alpha \times \mathcal{A}_{t+1}(r_t^{guess}) \text{ with } \alpha > 0$$

  - Decrease our new guess for r_t (i.e., increase the bond price) if there is excess demand for bonds ($\mathcal{A}_{t+1} > 0$)

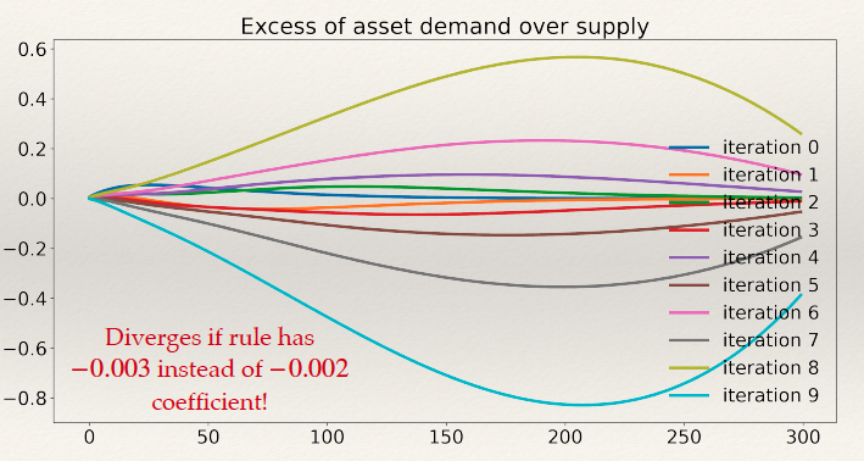
Introduction to SSJ

- The issue is that the previous updating rule may not work in practice...
- Results for $\alpha = -0.002$; algorithm converges after more than 100 iterations (very slow!)



Introduction to SSJ

- The issue is that the previous rule may not work in practice...
- Results for $\alpha = -0.003$; algorithm does not converge



Introduction to SSJ

- In more complicated models, with multiple markets and therefore multiple prices, updating the guesses can be much more challenging...
- **Here is when computing the "sequence-space Jacobian *J*" method can be useful**
  - Othe main benefit: it is fast!

Introduction to SSJ

- Suppose that we can compute, for each period $t \in [1, T]$, how the excess demand in period t changes with changes in r_s, for any $s \in [1, T]$
- That is, we are able to compute a $\boldsymbol{J}_{T \times T}$ Jacobian matrix, whose elements are given by:

$$J_{t,s} \equiv \frac{\partial \mathcal{A}_t(\boldsymbol{r}) - \bar{B}}{\partial r_{s+1}} = \frac{\partial \mathcal{A}_t\left(\{r_z\}_{z=1}^{T}\right)}{\partial r_{s+1}}$$

- If we know $\boldsymbol{J}$, we can then use a quasi-Newton method to update our guess for $\boldsymbol{r}$:

$$\left[\boldsymbol{r}^{update}\right]_{T \times 1} = \left[\boldsymbol{r}^{guess}\right]_{T \times 1} - \left[\boldsymbol{J}^{-1}\right]_{T \times T} \times \left[\boldsymbol{\mathcal{A}}\left(\boldsymbol{r}^{guess}\right) - \bar{\boldsymbol{B}}\right]_{T \times 1}$$

- This obviously simplifies our task regarding the rule that we use to update our guess for prices... and our algorithm will converge much more faster
  (remember that Newton methods have a quadratic convergence rate...)
- **The question is... how can we compute the $J_{t,s}$ elements?**

Introduction to SSJ

- We can apply in principle a **"Direct" or "Brute-Force" algorithm**
- Suppose that, for a given $\boldsymbol{r}^{guess}$, we are able to compute the dynamic path $\boldsymbol{\mathcal{A}}(\boldsymbol{r}^{guess})$.
- In order to compute $J_{t,s}$, we can slightly shock the interest rate at every $s \in [1, T]$ and see how $\mathcal{A}_t(.)$ reacts to it at every $t \in [1, T]$.
- With that reaction, we can then compute $J_{t,s}$ for every t and s as:

$$J_{t,s} = \frac{\mathcal{A}_t\left(\{r_z^{guess} + h_s\}_{t=1}^{T}\right) - \mathcal{A}_t\left(\{r_z^{guess}\}_{t=1}^{T}\right)}{(r_s^{guess} + h_s) - r_s^{guess}}$$

- For every $t = \{1, \ldots T\}$ of our "original" dynamic path, we need to solve for a "new" path, also of length T
- Hence, we have to apply our backward-forward iteration algorithm T times for each s
- In total, we have T^2 **backward-forward iterations** (which is computationally costly...)

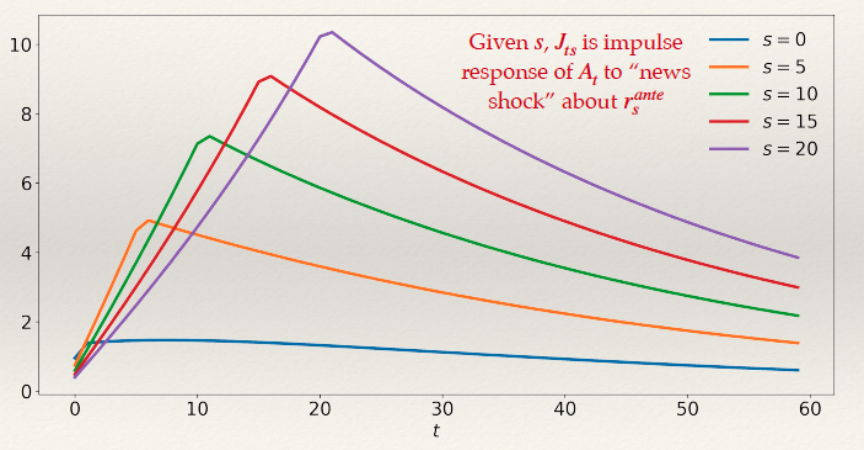
Introduction to SSJ

A small comment:

- We assume that the shock in period s is small enough so that the economy goes back to steady state after 1 period
- Otherwise, notice that at period $t = T - 1$, if the shock also happens at $s = T - 1$, then we cannot start our backward recursion at T from SS because the economy is not going to be at SS
- In that case, we would need to increase the length of the recursion for that particular s...

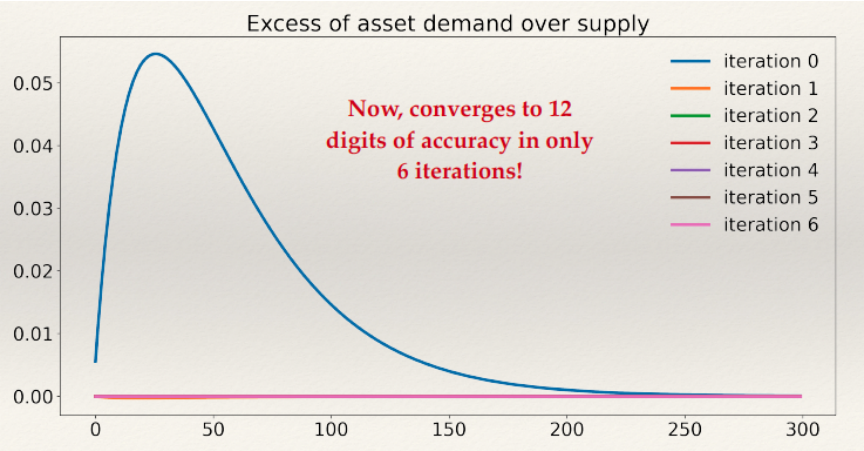
Introduction to SSJ

- This is how the Jacobian elements $J_{t,s}$ looks like $\rightarrow$ they are larger in magnitude whenever $t = s$ (i.e., shocks on impact)



Introduction to SSJ

- Apart from being computationally costly due to the T^2 backward-forward iterations, this method provides a great way to update our guess
- Just after 6 iterations, we solve for the entire $\boldsymbol{r}$ path



Introduction to SSJ

- The contribution behind the SSJ paper is to provide an algorithm to significantly speed up the previous process
- Notice that this algorithm is just to solve for an MIT shock. It is not an algorithm to solve directly for a model with aggregate uncertainty..
- However, once we have solved and computed paths for the MIT shocks, we can then apply the insight in the **BKM** paper
- Before going to the algorithm itself, I will re-write a version of the KS model (without agg. uncertainty) using the same notation as in the SSJ paper
  - The goal is to emphasize the dependence of policies on the path of prices

The Krusell-Smith Model (without Agg Uncertainty)

- The Bellman equation is:

$$
\begin{aligned}
V_t\left(e, k_{-}\right) &= \max_{c,k} u(c) + \beta \sum_{e'} V_{t+1}\left(e', k\right) P\left(e, e'\right) \\
\text{s.t.} c + k &= \left(1 + r_t\right) k_{-} + w_t e \times n \\
k &\geq 0
\end{aligned}
\tag{1}
$$

  - where n are the exogenous number of hours the household can work

- The production function is Cobb Douglas, $Y_t = Z_t K_{t-1}^{\alpha} N_t^{1-\alpha}$, where Z_t is the deterministic path for agg. productivity. The FOCs are:

$$
\begin{aligned}
r_t &= \alpha Z_t \left(\frac{K_{t-1}}{N_t}\right)^{\alpha-1} - \delta \\
w_t &= \left(1 - \alpha\right) Z_t \left(\frac{K_{t-1}}{N_t}\right)^{\alpha}
\end{aligned}
$$

The Krusell-Smith Model

- If we know the aggregate K_{t-1} we can pin down all current prices (since $N_t = \sum_e \pi(e) n$ is the exogenous labor supply).
- If we know the path for $\{K_t\}_{\forall t}$, we can then back out the path for all prices and we can solve the previous Bellman equation.
- The distribution, D, follows the following law of motion:

$$D_{t+1}\left(e', k\right) = \int_{k_-} \sum_{e'} \Lambda\left[\left(e, k_-\right), \left(e', k\right)\right] \times D_t\left(e, k_-\right) dk_-$$

  - where $D_t\left(e, k_-\right)$ denotes the measure of households in state e that own capital k_-beginning of period t.
- The transition matrix Λ is given by:

$$\Lambda\left[\left(e, k_-\right), \left(e', k\right)\right] = P\left(e, e'\right) \times \mathbb{I}_{\{k^\star(e,k_-)=k\}}$$

  - where $k_t^\star\left(e, k_-\right)$ denotes the individual policy function.

The Krusell-Smith Model

- The end-of-period demand for capital, that we denote as $\mathcal{K}_t$, is:

$$\mathcal{K}_t(.) = \sum_e \int_{k_-} k_t^\star(e, k_-) D_t(e, k_-) dk_- \tag{2}$$

- As we have studied in the previous lecture, the $k_t^\star(e, k_-)$ policy is a function of the current & future path $\{r_s, w_s\}_{s \geq t}$.
- A central part of the SSJ paper is to view endogenous agg. outcomes as function of prices

- To better illustrate this, we denote the end-of-period demand for capital as

$$\mathcal{K}_t\left(\{r_s, w_s\}_{s \geq 0}\right)$$

  - The sub-index is $s \geq 0$ (not $s \geq t$) since changes in prices even before t affect the current distribution D_t

The Krusell-Smith Model

- Together with the equations defining r_t and w_t, we can write the market clearing condition for the capital good market as:

$$
\begin{aligned}
H_t\left(\boldsymbol{K}, \boldsymbol{Z}\right) &= \mathcal{K}_t\left(\{r_s, w_s\}_{s\geq 0}\right) - K_t \qquad (3)\\
&= \mathcal{K}_t\left(\left\{\alpha Z_s\left(\frac{K_{s-1}}{\sum_e \pi(e)n}\right)^{\alpha-1} - \delta, (1-\alpha) Z_s\left(\frac{K_{s-1}}{\sum_e \pi(e)n}\right)^{\alpha}\right\}_{s\geq 0}\right) - K_t = 0
\end{aligned}
$$

  - where $\boldsymbol{K} = (K_0, K_1, \ldots)'$ and $\boldsymbol{Z} = (Z_0, Z_1, \ldots)'$ are vectors with the path for aggregate capital and productivity.
  - Given an initial K_{-1} and for the exogenous path $\boldsymbol{Z}$, Equation 3 pins down the equilibrium path of aggregate capital.

Impulse Response - 1st Order Approximation around the SS

- Suppose we have solved for the steady state.
- Consider some transitory shock to the aggregate productivity, $d\boldsymbol{Z} = (dZ_0, dZ_1, ..., 0, 0)'$
- We can directly apply the **implicit function theorem** to Equation 3 and get:

$$d\boldsymbol{K} = (-)\boldsymbol{H}_{\boldsymbol{K}}^{-1}\boldsymbol{H}_{\boldsymbol{Z}}d\boldsymbol{Z} \tag{4}$$

  - where $\boldsymbol{H}_{\boldsymbol{K}}$ and $\boldsymbol{H}_{\boldsymbol{Z}}$ are the Jacobians of $\boldsymbol{H}(.)$ with respect to $\boldsymbol{K}$ and $\boldsymbol{Z}$, evaluated at the steady state

- In practice, Equation 4 is solved up to a given (large) horizon T (allowing the economy to go back to SS, once the Z shocks fade away).

$$[d\boldsymbol{K}]_{T\times 1} = (-)\left[\boldsymbol{H}_{\boldsymbol{K}}^{-1}\right]_{T\times T}[\boldsymbol{H}_{\boldsymbol{Z}}]_{T\times T}[d\boldsymbol{Z}]_{T\times 1}$$

  - The backward+forward algorithm will give us $d\boldsymbol{Z}$ and $d\boldsymbol{K}$. But not the Jacobians...

Impulse Response - 1st Order Approximation around the SS

- We can use the **chain rule** to relate the Jacobians $[\boldsymbol{H_K}]_{T\times T}$ and $[\boldsymbol{H_Z}]_{T\times T}$ to the derivatives of $\mathcal{K}_t(.)$, evaluated at the steady state.
- **Chain rule:** After differentiating the previous elements with respect to K_s, the $\{t,s\}$ entry of $\boldsymbol{H_K}$ is:

$$[\boldsymbol{H_K}]_{\{t,s\}} = \frac{\partial \mathcal{K}_t(.)}{\partial r_{s+1}}\frac{\partial r_{s+1}}{\partial K_s} + \frac{\partial \mathcal{K}_t(.)}{\partial w_{s+1}}\frac{\partial w_{s+1}}{\partial K_s} - 1_{\{s=t\}}$$

$$[\boldsymbol{H_Z}]_{\{t,s\}} = \frac{\partial \mathcal{K}_t(.)}{\partial r_{s+1}}\frac{\partial r_{s+1}}{\partial Z_s} + \frac{\partial \mathcal{K}_t(.)}{\partial w_{s+1}}\frac{\partial w_{s+1}}{\partial Z_s}$$

- The derivatives $\left\{\frac{\partial r_{s+1}}{\partial K_s}, \frac{\partial w_{s+1}}{\partial K_s}, \frac{\partial r_{s+1}}{\partial Z_s}, \frac{\partial w_{s+1}}{\partial Z_s}\right\}$ can all be computed directly and evaluated at the steady state (given the closed-form expressions for r and w, based on the firm's FOCs)
- To compute $\boldsymbol{H_K}^{-1}\boldsymbol{H_Z}$, all we need are the Jacobians $\boldsymbol{J}$ of $\mathcal{K}$ wrt prices: $\frac{\partial \mathcal{K}_t(.)}{\partial r_{s+1}}$ and $\frac{\partial \mathcal{K}_t(.)}{\partial w_{s+1}}$

Brute Force Algorithm

- As we discussed before, we can compute $\frac{\partial \mathcal{K}_t(.)}{\partial r_{s+1}}$ and $\frac{\partial \mathcal{K}_t(.)}{\partial w_{s+1}}$ by **brute force**.
- That is, for every $\{t, s\} \in [1, T]$, we introduce a very small shock to the interest rate and we compute a whole new dynamic path of length T.
- For instance, for the simplified KS model with no endogenous labor supply (only r price):

$$J_{t,s} = \frac{\mathcal{K}_t\left(\{r_z^{guess} + h_s\}_{t=1}^T\right) - \mathcal{K}_t\left(\{r_z^{guess}\}_{t=1}^T\right)}{(r_s^{guess} + h_s) - r_s^{guess}}$$

- Notice that once we have $\boldsymbol{H}_{\boldsymbol{K}}^{-1}$ and $\boldsymbol{H}_{\boldsymbol{Z}}$ we have effectively solve a model with aggregate uncertainty, to the extent that aggregate responses to shocks are linear (this is the BKM insight)
- If I give you any other sequences of shocks, we can apply $d\boldsymbol{K} = (-)\boldsymbol{H}_{\boldsymbol{K}}^{-1}\boldsymbol{H}_{\boldsymbol{Z}} d\boldsymbol{Z}$ to get the response for capital

Comments and Link with BKM

- Notice that the idea is similar to that in the BKM paper that we saw before...
- They both rely on a linear approximation. That is, the size of the shocks and their interaction does not matter for aggregate responses (after re-scaling).
  - If second order moments are important, neither the BKM nor the SSJ would be useful...
- The insight of the SSJ paper is that we should compute the Jacobians to have a better rule on how we should update our guesses for the prices in our deterministic path... &, more importantly, they provide an algorithm that is very fast for computing those Jacobians.
- Before going to the algorithm itself, let's write down a **General Representation** of a model with heterogeneous agents

General Model Representation

- $\boldsymbol{X}_t$ is a vector of n_x inputs (typically, prices)
- $\boldsymbol{Y}_t$ is a vector of n_y aggregate outputs (or outcomes of the model).
- $\boldsymbol{D}_t$ is a vector with the distribution of agents (discretized in a fixed n_g number of points)
- $\boldsymbol{y}_t$ is a $n_g \times n_y$ matrix of individual outcomes
- If our aggregate output of interests are averages of individual outcomes:

$$[\boldsymbol{Y}_t]_{n_y \times 1} = [\boldsymbol{y}_t']_{n_y \times n_g} [\boldsymbol{D}_t]_{n_g \times n_y}$$

- Lastly, $\boldsymbol{v}_t$ denotes a n_x vector of value functions
- $\boldsymbol{\Lambda}$ is a transition matrix (that updates the distribution)

General Model Representation

- **Set of equations**: We assume that there exists 3 functions v, Λ, y, such that for a given initial distribution, $\boldsymbol{Y}_t$ is the solution to the system of equations exits:

$$\boldsymbol{v}_t = v\left(\boldsymbol{v}_{t+1}, \boldsymbol{X}_t\right) \tag{5}$$

$$\boldsymbol{D}_{t+1} = \Lambda\left(\boldsymbol{v}_{t+1}, \boldsymbol{X}_t\right)' \boldsymbol{D}_t \tag{6}$$

$$\boldsymbol{Y}_t = y\left(\boldsymbol{v}_{t+1}, \boldsymbol{X}_t\right)' \boldsymbol{D}_t \tag{7}$$

- Example for the Krusell-Smith model: $\boldsymbol{X}_t = (r_t, w_t)$; $\boldsymbol{y}_t = k(.)$; $\boldsymbol{Y}_t = (K_t, C_t)$

Steady state

- For a given $\boldsymbol{X}_{ss}$, the steady state of the model is the fixed point: $\left(\boldsymbol{Y}_{ss}, \boldsymbol{v}_{ss}, \boldsymbol{D}_{ss}\right)$ that obtains when $\boldsymbol{X}_t = \boldsymbol{X}_{ss}$ at all t:

$$\boldsymbol{v}_{ss} = v\left(\boldsymbol{v}_{ss}, \boldsymbol{X}_{ss}\right); \quad \boldsymbol{\Lambda}_{ss} \equiv \Lambda\left(\boldsymbol{v}_{ss}, \boldsymbol{X}_{ss}\right); \quad \boldsymbol{y}_{ss} \equiv y\left(\boldsymbol{v}_{ss}, \boldsymbol{X}_{ss}\right)$$

General Model Representation

Perturbation around the steady state (MIT shock)

- Consider transitions of length T that end at this steady state.
  - Terminal conditions are $\boldsymbol{X}_{T-1} = \boldsymbol{X}_{ss}$; $\boldsymbol{v}_T = \boldsymbol{v}_{ss}$; $\boldsymbol{D}_T = \boldsymbol{D}_{ss}$
  - Initial conditions: $\boldsymbol{X}_0 = \boldsymbol{X}_{ss}$; $\boldsymbol{v}_0 = \boldsymbol{v}_{ss}$; $\boldsymbol{D}_0 = \boldsymbol{D}_{ss}$
- Define a $h(.)$ **mapping** from the $T \times n_x$ stacked vector of inputs X (one for each period in the transition) to the $T \times n_y$ vector of stacked outputs Y as:

$$\boldsymbol{Y}_{[T \times n_y]} = h\left(\boldsymbol{X}_{[T \times n_x]}\right)$$

- We assume that all functions are differentiable around the steady state, which implies that $h(.)$ is differentiable
- Our goal is to characterize the **Jacobian** $\mathcal{J}$ of the h mapping evaluated at $\boldsymbol{X} = \boldsymbol{X}_{ss}$
- Next: Assume there is only 1 input and 1 output, $n_x = n_y = 1$

Some Notation...

- Define the $[e^s]_{T\times 1} = (0, ...1, ...0)$; vector with all 0's, expect in the sth entry
- For a given dx, we say there is a "shock at time s" when the $T \times 1$ input vector is given by: $\boldsymbol{X}^s = \boldsymbol{X}_{ss} + e^s dx$
- Let $\boldsymbol{v}_t^s, \boldsymbol{D}_t^s, \boldsymbol{Y}_t^s$ denote the solution to Equations 5-7, given $\boldsymbol{X}^s$.
- Let $\Lambda_t^s = \Lambda\left(\boldsymbol{v}_{t+1}^s, \boldsymbol{X}_t^s\right)$ denote the transition matrix between states at time t
- Let $\boldsymbol{y}_t^s = y\left(\boldsymbol{v}_{t+1}^s, \boldsymbol{X}_t^s\right)$ denote the individual outcome function at time t in response to the shock at time s
- Finally, denote with a d, the difference of all objects relative to their SS level.

$$
\begin{aligned}
dY_t^s &= Y_t^s - Y_{ss} \\
d\boldsymbol{y}_t^s &= \boldsymbol{y}_t^s - \boldsymbol{y}_{ss} \\
d\Lambda_t^s &= \Lambda_t^s - \Lambda_{ss} \\
d\boldsymbol{D}_t^s &= \boldsymbol{D}_t^s - \boldsymbol{D}_{ss}
\end{aligned}
$$

- The sth column of the Jacobian $\mathcal{J}$ is then: $\lim_{dx\to 0} \frac{d\boldsymbol{Y}^s}{dx}$

The Fake News Algorithm - Lemma 1

- Individual policies responses at period t to a shock at date s are the same as the responses at period $t-1$ to shock at $s-1$. All that matters is time until the shock

$$\boldsymbol{y}_t^s = \boldsymbol{y}_{t+k}^{s+k}$$

  - The proof follows from the recursive nature of $\boldsymbol{v}_t = \nu\left(\boldsymbol{v}_{t+1}, \boldsymbol{X}_t\right)$

- **Important implication:** For each t, we do not need to perform T backward iterations to compute $\boldsymbol{y}_t^s$, for every $s = 1, \ldots T$. We only need 1 iteration

- **Why?** Suppose that we compute $\boldsymbol{y}_t^{T-1}$ for every t (i.e., the change in policy response at t to a shock at $s = T-1$)
- Substituting for $k = T-1-s$, then:

$$\boldsymbol{y}_t^s = \boldsymbol{y}_{(T-1)-(s-t)}^{(T-1)} \text{ (for} t \leq s)$$

- If $t > s$, then $(T-1)-(s-t) \geq T$ and hence the economy is back at steady state; $\boldsymbol{y}_t^s = \boldsymbol{y}_{ss}$ (this is our truncation). This can lead to issues obviously... but we won't use those cases (as shown below)

The Fake News Algorithm - Lemma 1

Lemma 1: For any $s \geq 1$, $t \geq 1$:

$$\boldsymbol{y}_t^s = \begin{cases} \boldsymbol{y}_{ss} & s < t \\ \boldsymbol{y}_{T-1-(s-t)}^{T-1} & s \geq t \end{cases}$$

$$\boldsymbol{\Lambda}_t^s = \begin{cases} \Lambda_{ss} & s < t \\ \Lambda_{T-1-(s-t)}^{T-1} & s \geq t \end{cases}$$

- This economizes on backward iteration: we only have to iterate backward once from $s = T - 1$ to 0, not once for each column of Jacobian

The Fake News Algorithm - Lemma 1

- **Lemma 1** applies only to the backward iteration and for the individual policies...It does not work directly for aggregates
- **Why not?** Because the distribution is evolving in time
- To see this, define

$$\mathcal{F}_{t,s} \cdot dx \equiv dY_t^s - dY_{t-1}^{s-1},$$

  where $Y_t^s = [\boldsymbol{y}_t^s]' \times \boldsymbol{D}_t^s$ is an aggregate outcome. Notice that $\mathcal{F}_{t,s} = J_{t,s} - J_{t-1,s-1}$
- Based on Lemma 1, we may be tempted to argue that $\mathcal{F}_{t,s} = 0$
- That is not true because the Lemma 1 applies only to $\boldsymbol{y}_t^s$, but not to the distribution $\boldsymbol{D}_t^s$
- In fact, the distribution evolves over time and (in general) $\boldsymbol{D}_t^s \neq \boldsymbol{D}_{t-1}^{s-1}$

The Fake News Algorithm - Lemma 2 - Paper main insight

- **Lemma 2.** Assume that $D_0 = D_{ss}$. For an infinitesimal dx, and any $s \geq 1, t \geq 1$, then:

$$\begin{aligned}\mathcal{F}_{t,s} \cdot dx &\equiv dY_t^s - dY_{t-1}^{s-1} \\ &= [\boldsymbol{y}_{ss}]' \left[\Lambda'_{ss}\right]^{(t-1)} d\boldsymbol{D}_1^s\end{aligned}$$

- **Intuition**: $\mathcal{F}_{t,s}$ is the effect of anticipating at $t = 0$ the dx shock.
  - Suppose that we know the path of the aggregate output Y_t at all dates $t = 0, \ldots T-1$ in response to a shock at date $s-1$.
  - How does this compare to the path of Y_t at $t = 1, \ldots T-1$ in response to a shock at date s?
  - From Lemma 1, the individual change in policies is identical in the two cases. The only difference at the aggregate level comes from the difference in the initial distribution. In the second case is $\boldsymbol{D}_1^s$ rather than $\boldsymbol{D}_{ss}$.
  - Lemma 2 says that, up to a first order, this difference in the distribution affects aggregates at all dates as if agents followed their steady state behavior

The Fake News Algorithm - Lemma 2

- The $\mathcal{F}_{t,s}$ can be interpreted as the impulse response to a "**date-s fake news shock**":
  - a shock to date s announced at date 0, and retracted at date 1
- At date 0 agents react to the announcement, which leads to the $\boldsymbol{D}_1^s$ distribution. After the announcement is retracted, agents revert to their SS policies.
  - This is not exactly what's happening (the shock persists), but according to Lemma 2, it's a good first-order approximation of the evolution of aggregates
- Up to a first order, the effect on output at all dates $t \geq 1$ comes only from the change in the distribution the period just after the shock
  - We can "ignore" changes in individual policies... aggregate effects are not large...

The Fake News Algorithm - Lemma 3

- Using this Lemma we can compute all the elements of the Jacobian matrix recursively
- Remember that $\mathcal{F}_{t,s} = J_{t,s} - J_{t-1,s-1}$. Then:
  - For $s = 0$ or $t = 0$, set $J_{t,s} = \mathcal{F}_{t,s}$
  - For any other case: $J_{t,s} = \mathcal{F}_{t,s} + J_{t-1,s-1}$.
- **Definition: Expectation Vector**

$$\mathcal{E}_t = (\Lambda_{ss})^t \, y_{ss}$$

- **Lemma 3 (trivial)**: The expectation vector solves the recursion:

$$\mathcal{E}_t = \Lambda_{ss}\mathcal{E}_{t-1}; \text{with } \mathcal{E}_0 = \boldsymbol{y}_{ss}$$

- By directly combining Lemmas 1-3, we get the following proposition...

The Fake News Algorithm - Proposition 1

- **Proposition 1:** Assume that $\boldsymbol{D}_0 = \boldsymbol{D}_{ss}$. For infinitesimal dx, **define** the $\{t, s\}$ element of the fake news matrix $\mathcal{F}$ as:

$$\mathcal{F}_{t,s} \cdot dx = \begin{cases} dY_0^s & t = 0 \\ \mathcal{E}'_{t-1} dD_1^s & t \geq 1 \end{cases}$$

- where $dY_0^s = (d\boldsymbol{y}_0^s)' \boldsymbol{D}_{ss}$ and $d\boldsymbol{D}_1^s = (d\Lambda_0^s) \boldsymbol{D}_{ss}$.
- Then the Jacobian $\mathcal{J}$ of h satisfies the recursion $\mathcal{J}_{t,s} = \mathcal{J}_{t-1,s-1} + \mathcal{F}_{t,s}$ for $t, s \geq 1$ with $\mathcal{J}_{t,s} = \mathcal{F}_{t,s}$ for $t = 0$ or $s = 0$, and is therefore given by:

$$\mathcal{J}_{t,s} = \sum_{k=0}^{min\{s,t\}} \mathcal{F}_{t,k,s-k}$$

- Reminder: Once we have the Jacobians we have effectively solved the model!

The Fake News Algorithm - Steps of the Algorithm

1. Compute $\boldsymbol{y}_0^s$ and Λ_0^s for every $s \in [0, T-1]$ using just one backward iteration from $T-1$.
   - these are the period-0 response to a shock in period s
   - we only need to do this for one shock at $T-1$, since $\boldsymbol{y}_0^s = \boldsymbol{y}_{T-1-s}^{T-1}$ and $\Lambda_0^s = \Lambda_{T-1-s}^{T-1}$
   - use Young method to compute the transition matrix Λ

- Using the steady-state distribution, for each s compute

$$\mathcal{Y}_s dx \equiv dY_0^s = \left(d\boldsymbol{y}_0^s\right)' \boldsymbol{D}_{ss} = \left(\boldsymbol{y}_0^s - \boldsymbol{y}_{ss}\right)' \boldsymbol{D}_{ss}$$

   - (captures the effect on the agg output at date 0 from the shock at date s)

$$\mathcal{D}_s dx \equiv d\boldsymbol{D}_1^s = \left(d\Lambda_0^s\right)' \boldsymbol{D}_{ss} = \left(\Lambda_0^s - \Lambda_{ss}\right)' \boldsymbol{D}_{ss}$$

   - (captures the change in the distribution at date 1 from the shock at date s)

2. Compute, for every t, $\mathcal{E}_t$ using the recursion

$$\mathcal{E}_t = \Lambda_{ss}\mathcal{E}_{t-1}; \text{ with } \mathcal{E}_0 = \boldsymbol{y}_{ss}$$

The Fake News Algorithm - Steps of the Algorithm

3 Based on Proposition 1, $\mathcal{F}_{t,s} \cdot dx = \begin{cases} dY_0^s & t = 0 \\ \mathcal{E}'_{t-1} dD_1^s & t \geq 1 \end{cases}$, compute the fake news matrix:

$$\mathcal{F} = \begin{bmatrix} \mathcal{Y}_0 & \mathcal{Y}_1 & \mathcal{Y}_2 & \dots & \mathcal{Y}_{T-1} \\ \mathcal{E}'_0 \mathcal{D}_0 & \mathcal{E}'_0 \mathcal{D}_1 & \mathcal{E}'_0 \mathcal{D}_2 & \dots & \mathcal{E}'_0 \mathcal{D}_{T-1} \\ \mathcal{E}'_1 \mathcal{D}_0 & \mathcal{E}'_1 \mathcal{D}_1 & \mathcal{E}'_1 \mathcal{D}_2 & \dots & \mathcal{E}'_1 \mathcal{D}_{T-1} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \mathcal{E}'_{T-2} \mathcal{D}_0 & \mathcal{E}'_{T-2} \mathcal{D}_1 & \mathcal{E}'_{T-2} \mathcal{D}_2 & \dots & \mathcal{E}'_{T-2} \mathcal{D}_{T-1} \end{bmatrix}$$

4 Use the previous matrix to compute the Jacobian recursively:

$$\mathcal{J}_{t,s} = \sum_{k=0}^{min\{s,t\}} \mathcal{F}_{t,k,s-k}$$

- For instance, $\mathcal{J}_{2,2} = \mathcal{E}'_1 \mathcal{D}_2 + \mathcal{E}'_0 \mathcal{D}_1 + \mathcal{Y}_0$

- Once we have the full Jacobian matrix $\mathcal{J}_{T \times T}$ we have solved the model. In the KS example, we can now compute $d\boldsymbol{K} = (-)\boldsymbol{H}_{\boldsymbol{K}}^{-1} \boldsymbol{H}_{\boldsymbol{Z}} d\boldsymbol{Z}$ and we are done

The Fake News Algorithm - Steps of the Algorithm

Table 2: Direct and fake news algorithms to compute 300×300 Jacobians $\mathcal{J}$.

	Krusell-Smith	HD Krusell-Smith	one-asset HANK	two-asset HANK
Direct	**21 s**	**2102 s**	**156 s**	**956 s**
step 1 (backward)	13 s	1302 s	132 s	846 s
step 2 (forward)	8 s	800 s	24 s	111 s
Fake news	**0.086 s**	**10.467 s**	**0.317 s**	**3.498 s**
step 1	0.060 s	8.654 s	0.236 s	3.159 s
step 2	0.011 s	1.061 s	0.022 s	0.119 s
step 3	0.011 s	0.758 s	0.045 s	0.201 s
step 4	0.003 s	0.003 s	0.014 s	0.018 s
Grid points n_g	3,500	250,000	3,500	10,500
Inputs n_x	2	2	4	5
Outputs n_y	2	2	4	4
Jacobians $n_x \times n_y$	4	4	16	20

Resources

- https://mhaense1.github.io/SSJ_Julia_Notebook/SSJ_notebook.html
- https://github.com/nikolakou/RA_HetAgents

Agent Heterogeneity: Higher-order Moments and Aggregate Non-linearities

Discussion

- Both the BKM and the SSJ algorithms are based on a 1st-order linear approximation
  - While they preserve all the non-linearities at the agent level, they do require a linear approximation for aggregates...
- We cannot use those model if higher-order aggregate moments are an important element of our model...
  - For instance, models with uncertainty or risk premia; models for which we care about aggregate precautionary savings motive...
- Similarly, we cannot use those algorithms (at least, without making further assumptions) if there are occasionally binding aggregate constraints
  - Models that feature a zero lower bound (ZLB); models with financial frictions; models in which one large agent can default...
- In these cases, we can use the KS algorithm, adding the relevant aggregate states to capture these higher-order moments... Or we can use other algorithms that do not rely on a 1st order perturbation

List of Algorithms we Will Cover

- Fernandez-Villaverde, Hurtado, Nuño. Financial Frictions and the Wealth Distribution (2023, Econometrica)
- Algan, Allais, Den Haan. Solving heterogeneous-agent models with parameterized cross-sectional distributions (2008; Jouranl of Economic Dynamics and Control)

Fernandez-Villaverde, Hurtado, Nuño

Fernandez-Villaverde, Hurtado, Nuño

- The algorithm is a twist of the KS algorithm
- We view the problem in the state space (not the sequence space)
- The depart from KS is that they use a highly-non linear perceived law of motion (PLM) for the aggregates
- In KS, remember that we guess a PLM of the type

$$\log K' = a_0 + a_1 \log K \text{ if } z = z_g$$
$$\log K' = b_0 + b_1 \log K \text{ if } z = z_b$$

  - That PLM was "good enough" in this model because households' consumption (saving) decision rule is close to linear with respect households' wealth
- In more complicated models, this may not be the case and we would need a more complicated PLM...
- The question is how can we find a "good" PLM (remember, I mentioned this is sometimes an "art")
- In this paper, the authors use a neural network to find the PLM as a nonlinear functions of the aggregate states

Fernandez-Villaverde, Hurtado, Nuño

- The steps of the algorithm are analogous to those in KS. The only difference is at the simulation stage...
  - After simulating the model, instead of running OLS... the authors employ a neural network to update their PLM, $\hat{H}$
- **Steps**
  1. Write down the problem recursively; define the relevant idiosyncratic, $\boldsymbol{s}$, and (bounded) aggregate states, $\boldsymbol{S}$.
     - ★ Notice that we still need to pick the relevant aggregate variables. There is still trial and error in this step... The aggregate states that we include should be enough to summarize the (relevant aspects of) distribution...
  2. Guess a perceived law of motion $\boldsymbol{S}' = \hat{H}(\boldsymbol{S})$.
     - ★ We have full flexibility here... Our guess doesn't have to be linear
  3. Using $\hat{H}$, we can solve the agents' Bellman equation
  4. Simulate the model (preferably, use Young method)
  5. Use a **neural network** to obtain a new guess for $\hat{H}$. Repeat until convergence of $\hat{H}$

Fernandez-Villaverde, Hurtado, Nuño

- What is a neural network? How can we use it in this context?
- Suppose our aggregate state space is $\boldsymbol{S} = \{S^1, S^2, ...S^I\}$
- Let $\boldsymbol{x} = \{x^1, x^2, ...x^I\}$ be the simulated data (aggregates only).
  - Each x^i is a $T \times 1$ vector and represents the simulated data for each agg variable
- In this paper, they used a neural network with 1-hidden layer and Q "neurons" as:

$$H\left(\underbrace{\boldsymbol{x}}_{\text{Data}}; \underbrace{\boldsymbol{\theta}}_{\text{Parameters}}\right) = \theta_0^{(2)} + \sum_{q=1}^{Q} \theta_q^{(2)} \phi\left(\theta_{0,q}^{(1)} + \sum_{i=1}^{|\boldsymbol{x}|} \theta_{i,q}^{(1)} x^i\right)$$

- A NN with 2-hidden layers looks like this:

$$H\left(\underbrace{\boldsymbol{x}}_{\text{Data}}; \underbrace{\boldsymbol{\theta}}_{\text{Parameters}}\right) = \theta_0^{(3)} + \sum_{r=1}^{R} \theta_r^{(3)} \phi\left(\theta_{0,r}^{(2)} + \sum_{q=1}^{Q} \theta_{q,r}^{(2)} \phi\left(\theta_{0,q}^{(1)} + \sum_{i=1}^{|\boldsymbol{x}|} \theta_{i,q}^{(1)} x_i\right)\right)$$

Fernandez-Villaverde, Hurtado, Nuño

$$H\left(\underbrace{\boldsymbol{x}}_{\text{Data}};\ \underbrace{\boldsymbol{\theta}}_{\text{Parameters}}\right) = \theta_0^{(2)} + \sum_{q=1}^{Q} \theta_q^{(2)} \phi\left(\theta_{0,q}^{(1)} + \sum_{i=1}^{|\boldsymbol{x}|} \theta_{i,q}^{(1)} x^i\right)$$

How it works? A neural network is a chain of "geometric transformations"

- We first apply a linear transformation to the simulated data
- Then, we twist this linear transformation through an "activation" function, $\phi(x)$
- We repeat this process in each layer of the network
- Notice that if $\phi(x) = x$, the we have a linear specification (OLS...)
- In practice, $\phi(x)$ is a Sigmoid function, $\phi(x) = \frac{1}{1+e^{-x}}$; Relu, $\phi(x) = \max\{0, x\}$; Softplus, $\phi(x) = \log(1 + e^x)$;...

Fernandez-Villaverde, Hurtado, Nuño

- Given our simulated data (length T), the network weights θ are trained by minimizing a quadratic error function:

$$\boldsymbol{\theta}^{\star} = argmin_{\boldsymbol{\theta}} \frac{1}{2} \sum_{j=1}^{T} || \, h\left(\boldsymbol{x}_j, \boldsymbol{\theta}\right) - \boldsymbol{x}_j' \, ||$$

- There are many algorithms to solve the previous problem (Gradient descent...)
- Careful in defining the "training sample"
  - If you train your model on the entire dataset, the network might learn the data "too well", memorizing specific examples rather than learning general patterns.
  - This is called **overfitting** $\rightarrow$ the model will perform well on the training data but poorly on unseen data because it hasn't learned to generalize
  - Without splitting the data into training and test (or validation) sets, you have no way to evaluate how well the model will perform on unseen data
- **Resources**:
  - **Matlab code**: `https://github.com/jesusfv/financial-frictions?tab=readme-ov-file`
  - **Julia packages**: `https://juliapackages.com/c/neural-networks`

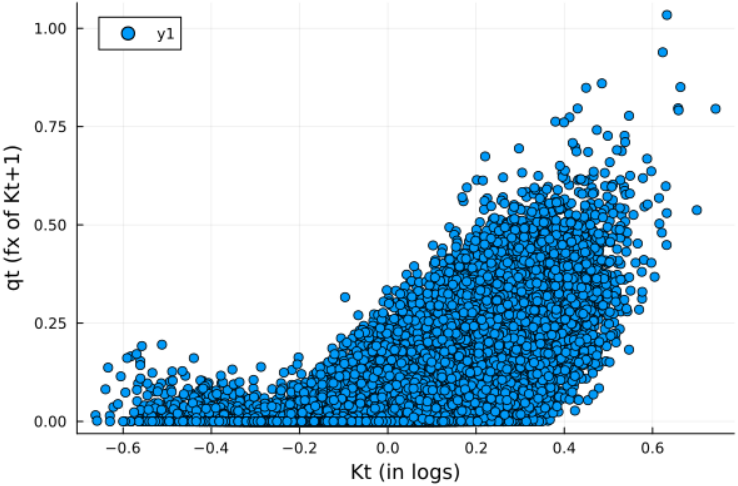
Fernandez-Villaverde, Hurtado, Nuño

- Let's consider a simple example
- Suppose we have 1 endogenous aggregate variable, K, and one endogenous price, q. Agents need to have a conjecture for current and future q in order to solve their problem
- Suppose that the true LOM (unknown to us) is the following:

$$\log K_{t+1} = a + b \log K_t + \sigma \epsilon_t$$
$$q_t = c \log K_{t+1} + d \log K_{t+1}^2$$

Fernandez-Villaverde, Hurtado, Nuño

- If d is large enough... we get the following non-linear relation



- Since this is non-linear, will a NN outperform OLS?

Fernandez-Villaverde, Hurtado, Nuño

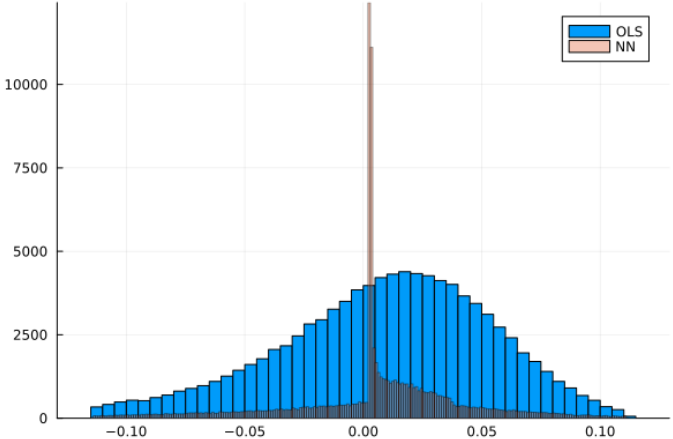
- In the following example, I first simulate data for $\{K, q\}$ based on the "true" process defined above
- Using the simulated data, I run **OLS** to estimate c and d

$$q_t = c \log K_{t+1} + d \log K_{t+1}^2 + error_t$$

  - I use those estimates to predict $\hat{q}_t^{OLS}$.
  - I then compute the prediction error: $MSE^{OLS} = \frac{1}{T} \sum_{t=1}^{T} \left(\hat{q}_t^{OLS} - q_t\right)^2$
- We repeat similar steps for the **neural network**
  - A detailed explanation on how we can estimate ("train") a NN is outside the scope of this class...
  - But there are many packages in Julia that do this for us very easily! For instance, **Flux.jl**
  - We just need to define how many "layers" we want; the type of activation functions ϕ's, and the algorithm that the **Flux.jl** will use to estimate the θ's

Fernandez-Villaverde, Hurtado, Nuño

- Here, we compare $\{(\hat{q}_t^{OLS} - q_t)^2\}_{\forall t}$ vs $\{(\hat{q}_t^{NN} - q_t)^2\}_{\forall t}$



- In this example, using a NN significantly decreases magnitudes of the errors!
- This may not always be the case... It depends on our problem at hand...

Algan, Allais, Den Haan

Algan et al. 2008

- In the KS algorithm (and in the Fernandez-Villaverde et al. paper), we have summarized the distribution using a finite set of moments $\boldsymbol{m}$
  - We use those moments $\boldsymbol{m}$ as states when writing down the agents' recursive problem and we guess a conjectured LOM for $\boldsymbol{m}'$
  - The "hope" in those methods is that the information in $\boldsymbol{m}$ and the conjectured $\boldsymbol{m}$' is enough to accurately pin down the future evolution of prices (bounded rationality assumption)
- In **Algan et al. 2008**, the idea is to directly parameterized the agent distribution
  - Their method involves having a flexible specification to approximate the density $g(.)$ (or distribution) of agents

Algan et al. 2008

- Suppose that we solve for the **SS of the model** and we compute the invariant distribution, $g(.)$
- We can then compute a set of **"reference moments"** $\widetilde{\boldsymbol{m}}$ based on the SS distribution, which could capture higher-order moments of the distribution.
  - For instance, variance, kurtosis, even some percentiles,...
- Algan et al. use those SS reference moments to approximate the density $g(.)$ that characterizes the distribution
- Moving to the **model with aggregate uncertainty**, we then define a set of aggregate moments $\boldsymbol{m}$ (a subset of $\widetilde{\boldsymbol{m}}$) that will be part of our state space
- Based on $\boldsymbol{m}$ and on the information contained in the SS reference moments ($\widetilde{\boldsymbol{m}}$), we directly parameterized the distribution of agents

Algan et al. 2008

- Consider a model in which agents have 2 endogenous idiosyncratic states, $\{\epsilon, k\}$
- We solve for the steady state and compute the density $g(\epsilon, k)$
- We then approximate that density as follows:

$$g(\epsilon, k) \approx g_0 \times \exp\left\{ g_1^1 \left(\epsilon - \widetilde{m}_1^1\right) + g_1^2 \left(k - \widetilde{m}_1^2\right) + \sum_{i=2}^{n_g} \sum_{j=0}^{i} g_i^j \left(\left[\epsilon - \widetilde{m}_1^1\right]^{i-j} \left[k - \widetilde{m}_1^2\right]^j - \widetilde{m}_i^j \right) \right\}$$

  - n_g denotes the degree of approximation (1st order, 2nd order,..)
  - $\widetilde{m}_1^1$ and $\widetilde{m}_1^2$ are the first moments of the distribution for ϵ and k, respectively (means)
  - $\{\widetilde{m}_i^j\}_{i,j=(2,0)}^{(n_g,i)}$ are centralized higher-order moments

Algan et al. 2008

- For instance if we only include first moment (mean) of the distribution:

$$g(\epsilon, k) \approx g_0 \times \exp\left\{g_1^1\left(\epsilon - \tilde{m}_1^1\right) + g_1^2\left(k - \tilde{m}_1^2\right)\right\}$$

- If we only include first two moments (mean and variance) of the distribution:

$$g(\epsilon, k) \approx g_0 \times \exp\left\{g_1^1\left(\epsilon - \tilde{m}_1^1\right) + g_1^2\left(k - \tilde{m}_1^2\right) + \sum_{j=0}^{2} g_2^j\left(\left[\epsilon - \tilde{m}_1^1\right]^{2-j}\left[k - \tilde{m}_1^2\right]^j - \tilde{m}_2^j\right)\right\}$$

- Hence:

$$g(\epsilon, k) \approx g_0 \times \exp\left\{g_1^1\left(\epsilon - \tilde{m}_1^1\right) + g_1^2\left(k - \tilde{m}_1^2\right) + g_2^0\left(\left[\epsilon - \tilde{m}_1^1\right]^2 - \tilde{m}_2^1\right) + g_2^2\left(\left[k - \tilde{m}_1^2\right]^2 - \tilde{m}_2^2\right) + \right.$$
$$\left. + g_2^1\left(\left[\epsilon - \tilde{m}_1^1\right]\left[k - \tilde{m}_1^2\right] - \tilde{m}_2^1\right)\right\}$$

Algan et al. 2008

- To solve for the coefficients, g, we need to make sure that our parameterized density is consistent with the set of reference moments
- That is:

$$\widetilde{m}_1^1 = \int\int \epsilon \times g(\epsilon, k) d\epsilon dk$$
$$\widetilde{m}_1^2 = \int\int k \times g(\epsilon, k) d\epsilon dk$$
$$\widetilde{m}_i^j = \int\int \left(\epsilon - \widetilde{m}_1^1\right)^{i-j} \left(k - \widetilde{m}_1^2\right)^j g\left(\epsilon, k\right) d\epsilon dk$$

Algan et al. 2008

- This is where the particular functional form in Algan et al. comes in handy.
- Notcei that we can solve for the vector of parameters $\boldsymbol{g}$ by solving this minimization problem:

$$\min_{\boldsymbol{g}} \int\int g\left(\epsilon, k\right) d\epsilon dk$$

- **Why?** Because we have that the FOCs wrt to the parameters are:

$$\begin{aligned}
\partial \frac{\int\int g\left(\epsilon, k\right) d\epsilon dk}{\partial g_1^1} &= \\
&= \int\int \partial \frac{g\left(\epsilon, k\right)}{\partial g_1^1} \; d\epsilon dk \\
&= \int\int \; g\left(\epsilon, k\right) \times \left(\epsilon - \widetilde{m}_1^1\right) \; d\epsilon dk \\
&= 0
\end{aligned}$$

- The same applies for every other moment...

Algan et al. 2008

- When moving to the model with aggregate uncertainty, we do not need to keep track of all the reference moments as state variables
- Instead, we can define a subset of moments $\boldsymbol{m}$ that we will keep track of (for instance, first-order moments only)
- In our example, we can include as states only $m_1^1 = \mathcal{E}$ and $m_1^2 = K$
- As for the reference moments, we use their SS values

$$g(\epsilon, k; \mathcal{E}, K) \approx g_0 \times \exp\Bigg\{ \underbrace{g_1^1(\epsilon - \mathcal{E}) + g_1^2(k - K)}_{\text{Means}} + \underbrace{g_2^0\left([\epsilon - \mathcal{E}]^2 - \widetilde{m}_2^1\right) + g_2^2\left([k - K]^2 - \widetilde{m}_2^2\right)}_{\text{Variances}} + \underbrace{g_2^1\left([\epsilon - \mathcal{E}][k - K] - \widetilde{m}_2^1\right)}_{\text{Covariance}} \Bigg\}$$

- Here, the coefficients g_i^j are going to depend on $\{\mathcal{E}, K\}$. Once we have a grid for those values, we would need to solve for g_i^j for every combination of $\{\mathcal{E}, K\}$

Algan et al. 2008

- **Recap**: While we only track the evolution of aggregate $\{\mathcal{E}, K\}$, we have parameterized the full distribution based on some information from the steady state.
- By doing this, we basically known the entire distribution of agents
- Now, let's see the full algorithm...

Algan et al. 2008

- Suppose we only use $\{\mathcal{E}, K\}$ as aggregate moments
- Define a grid for $\mathcal{E}$ and K. For each combination of $\{\mathcal{E}, K\}$, solve for the coefficients g_i^j, based on some vector of steady state reference moments $\widetilde{\boldsymbol{m}}$
- Guess a PLM for $\mathcal{E}'$ and K'. Here the PLM can be fully non-linear
  - With this information, we are able to compute the current and the next-period distribution (hence, we can compute any price we want given some policies) and we can solve for the agents' problem
- Guess the agents' value function and solve for their optimal policies $\{\epsilon'(.), k'(.)\}$
- Use those policies and the parameterized distribution to compute

$$\mathcal{E}'(.) = \int\int g(\epsilon, k) \times \epsilon'(\epsilon, k) \; d\epsilon dk$$

$$K'(.) = \int\int g(\epsilon, k) \times k'(\epsilon, k) \; d\epsilon dk$$

- Check that these aggregate outcomes are consistent with the guessed PLM. If not, update the PLM accordingly; solve again for policies. Keep iterating until convergence of the PLM
- Continue with this iteration until convergence of the value function

Algan et al. 2008

- **Key benefit of Algan et al.**: We do not need to simulate the model to find a solution!
  - In KS, the simulation stage can be time consuming... especially if we need to solve for a (non-trivial) market clearing condition in each period..
- **Possible refinement:**
  - We may update at some point the reference moments, as a function of $\mathcal{E}$ and K
  - For this, we do need simulations
  - Basically, we follow the same algorithm as before... but after convergence of the value function we simulate the model.
  - We compute by simulation the reference moments of relevance. For instance, the cross-sectional variance of capital, m_2^2. We can then run an OLS specification such that: $m_2^2 = a_0 + a_1 K + a_2 \mathcal{E} + error$ and update the coefficients